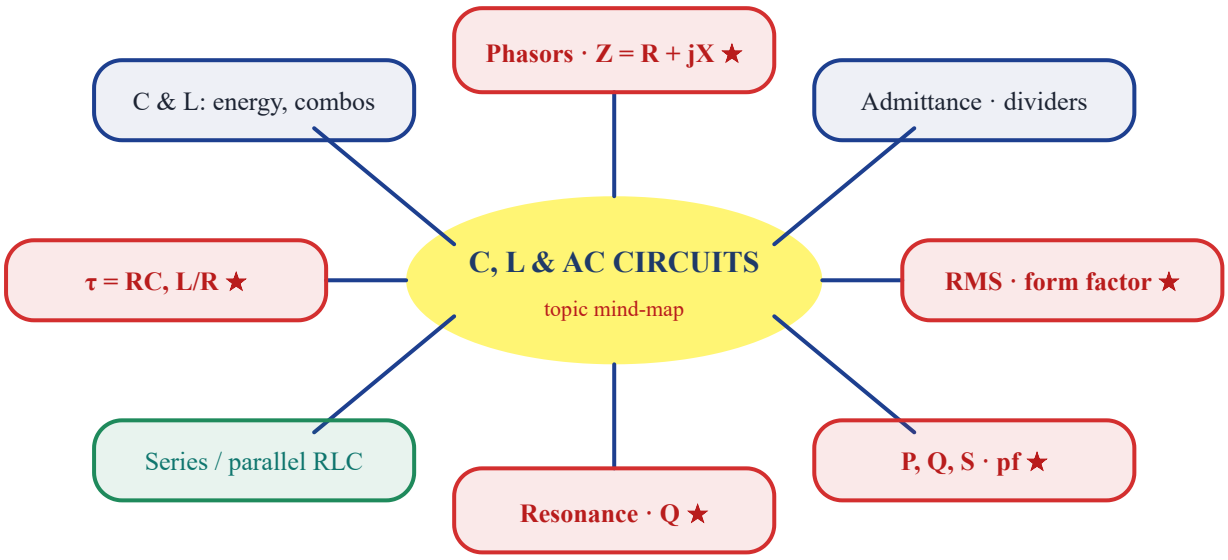
Capacitors, Inductors & AC Circuits

C & L · transients · RMS · phasors · impedance · resonance · AC power: exam-ready in 11 sheets



↑ the whole topic on one page: red = high-yield, green = method / applied

Syllabus coverage

"Capacitors and inductors: inductors and capacitors, their characteristics, **series-parallel combination** of inductors and capacitors; **RLC transients**. Series and parallel AC circuits: **impedance and phasor diagram**, series and parallel networks, voltage divider rule, **admittance and susceptance**". Sinusoid values, resonance and AC power are added because they are tested with this topic. AC mesh/nodal analysis, Y–Δ and the theorems reuse Lesson 2's methods with Z in place of R.

High-yield list

① ½CV², ½LI²
② τ = RC, L/R (63.2%)
③ 0.707 V_m, FF 1.11
④ ELI the ICE man
⑤ $f_0 = 1/(2\pi\sqrt{LC})$
⑥ P = VI cos φ

Legend: **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

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1 Capacitor & Inductor: the Storage Elements

Capacitor: Two conductors separated by a dielectric. Stores energy in the **electric field**. C = Q/V, unit **farad** (1 F = 1 C/V). Parallel plates: **C = εA/d**.

Inductor: A coil of wire. Stores energy in the **magnetic field**. L = NΦ/I, unit **henry** (1 H = 1 Wb/A = 1 V·s/A). Solenoid: **L = μN²A/l**.



E-field: W = ½CV² B-field: W = ½LI²

C keeps charge on its plates; L keeps flux in its core

Property	Capacitor C	Inductor L
Defining law	**i = C dv/dt**	**v = L di/dt**
Integral form	v = (1/C)∫i dt	i = (1/L)∫v dt
Energy stored	**½CV² = Q²/2C = ½QV**	**½LI²**
Cannot change instantly	**voltage**	**current**
DC steady state	open circuit	short circuit
At switch-on (no initial energy)	short circuit	open circuit
AC phase	I **leads** V by 90°	I **lags** V by 90°
Series	$1/C = \Sigma\, 1/C_k$	$L = \Sigma\, L_k$
Parallel	$C = \Sigma\, C_k$	$1/L = \Sigma\, 1/L_k$

Memory trick: **L behaves like R** (series adds). **C is R upside-down** (parallel adds). C and L are **duals**: swap v ↔ i and C ↔ L to turn one row into the other.

Worked example ✍️: 2 μF, 3 μF, 6 μF on 100 V. **Series:** 1/C = ½ + ⅓ + ⅙ = 1 → **C = 1 μF**; Q = CV = 100 μC on **each**; V = Q/C → 50 V, 33.3 V, 16.7 V (sum 100 ✓); W = ½ × 1 μ × 100² = **5 mJ**. **Parallel:** C = 11 μF, W = 55 mJ.

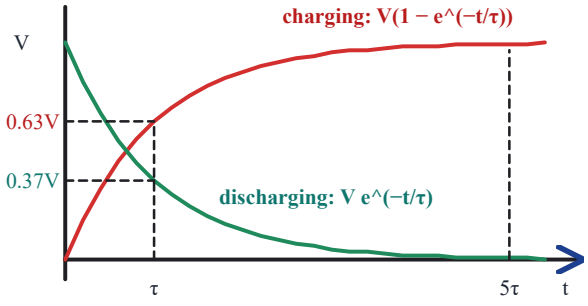
Inductors 4 H ‖ 12 H (no mutual coupling) = 48/16 = **3 H**; at 2 A, W = ½ × 3 × 2² = **6 J**.

Exam trap: Capacitors in series carry the **same charge**, so the **smallest C takes the largest voltage** (2 μF gets 50 V). Series C is always smaller than the smallest one. Ideal C and L dissipate **no power**: they store energy and give it back.

2 RC & RL Transients (First Order)

Transient: The change from one steady state to another after a switch operates. In a circuit with one C or one L it is an **exponential** set by the time constant τ: **$\tau = RC$** or **$\tau = L/R$** (R = Thevenin resistance seen by C or L).

One formula for every case: $x(t) = x(\infty) + [x(0^+) - x(\infty)]\, e^{-t/\tau}$ (x = v_C or i_L)



t	charging	decay left
1τ	**63.2%**	**36.8%**
2τ	86.5%	13.5%
3τ	95.0%	5.0%
4τ	98.2%	1.8%
5τ	99.3%	0.7%

RC charge: $v_C = V(1 - e^{-t/RC})$

RC discharge: $v_C = V_0 e^{-t/RC}$

RL rise: $i = (V/R)(1 - e^{-Rt/L})$

RL decay: $i = I_0 e^{-Rt/L}$

Worked example: RC ✍️: 10 kΩ, 100 μF, 12 V: τ = **1 s**; $i(0^+)$ = 1.2 mA; $v_C(1\text{ s}) = 12 \times 0.632 =$ **7.59 V**; 6 V at $t = \tau \ln 2 =$ **0.693 s**

Worked example: RL ✍️: 2 H, 4 Ω, 12 V: τ = L/R = **0.5 s**; $i(\infty)$ = 3 A; $i(0.5\text{ s}) = 3 \times 0.632 =$ **1.90 A**; final energy $\frac{1}{2} \times 2 \times 3^2 =$ **9 J**

Memory trick: **63 – 86 – 95 – 98 – 99**: one number per τ. "**Five taus and it's done.**"

Exam trap: RL: τ = **L/R**, not LR. A bigger R makes RL **faster** but RC **slower**. i_C and v_L decay ($e^{-t/\tau}$).

3 Second-Order (RLC) Response

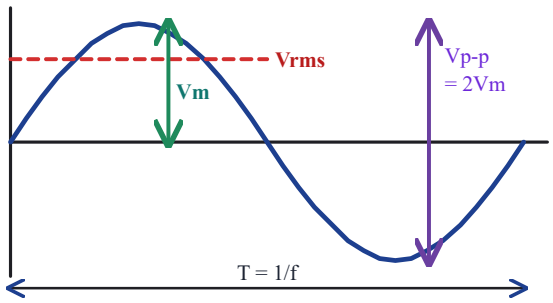
$\omega_0 = 1/\sqrt{LC}$ (undamped natural frequency) · damping factor **$\alpha = R/2L$ (series), $\alpha = 1/(2RC)$ (parallel)** · $\zeta = \alpha/\omega_0$ · damped frequency $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$

Condition	Response	Series RLC
$\alpha > \omega_0$ ($\zeta > 1$)	overdamped: two exponentials, slow	$R > 2\sqrt{L/C}$
$\alpha = \omega_0$ ($\zeta = 1$)	critically damped: **fastest** without oscillation	$R = 2\sqrt{L/C}$
$\alpha < \omega_0$ ($\zeta < 1$)	underdamped: decaying oscillation at ω_d	$R < 2\sqrt{L/C}$

e.g. L = 1 H, C = 1 μF → ω_0 = 1000 rad/s; critical R (series) = $2\sqrt{10^6}$ = **2 kΩ**.

4 Sinusoids: Peak, RMS, Average

Sinusoid: $v(t) = V_m \sin(\omega t + \varphi)$: V_m = amplitude (peak), **$\omega = 2\pi f$** rad/s, $T = 1/f$, φ = phase angle. Bangladesh mains: **220 V rms, 50 Hz**.



RMS: The DC value that gives the **same heating** in a resistor: $V_{rms} = \sqrt{(\text{mean of } v^2)}$. Meters and ratings ("220 V") are RMS.

Average: Mean over a **half cycle**: $2V_m/\pi$. Over a full cycle a pure sine averages **0**.

Waveform (peak V_m)	V_{rms}	V_{avg}	FF = rms/avg	Crest = peak/rms
Sine / full-wave rectified	**0.707 V_m**	**0.637 V_m**	**1.11**	**1.414**
Half-wave rectified sine	$V_m/2$	V_m/π	1.57	2
Square ($\pm V_m$)	V_m	V_m	1	1
Triangle / sawtooth	$V_m/\sqrt{3}$	$V_m/2$	1.155	$\sqrt{3} = 1.732$

Worked example 1 ✍️: BD mains 220 V, 50 Hz: $V_m = 220\sqrt{2} =$ **311 V**; $V_{p\text{-}p} = 622$ V; $\omega = 2\pi \times 50 =$ **314 rad/s**; $T =$ **20 ms**; $V_{avg} = 0.637 \times 311 = 198$ V.

Worked example 2 ✍️: $v = 100 \sin(314t + 30°)$ V: $V_m = 100$ V, $V_{rms} =$ **70.7 V**, $f = 314/2\pi \approx$ **50 Hz**, $\varphi = +30°$; at $t = 0$, $v = 100 \sin 30° =$ **50 V**.

Phase difference

$v = 10 \sin(\omega t + 20°)$, $i = 2 \sin(\omega t - 40°)$ → v **leads** i by **20° − (−40°) = 60°**. Write both as sin with positive amplitude first: $\cos \omega t = \sin(\omega t + 90°)$, $-\sin \omega t = \sin(\omega t + 180°)$.

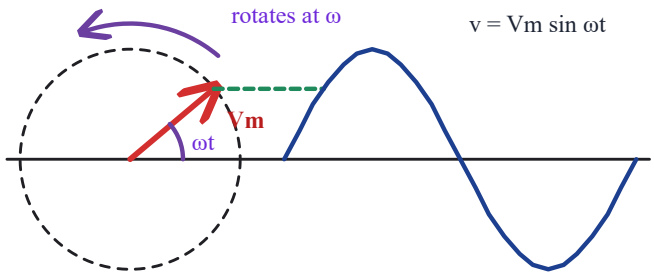
Derivation (written) ✍️: $V_{rms}^2 = (1/T)\int_0^T V_m^2 \sin^2\omega t \, dt = (V_m^2/T)\int_0^T \tfrac{1}{2}(1 - \cos 2\omega t) \, dt = V_m^2/2 \rightarrow$ **$V_{rms} = V_m/\sqrt{2}$**. Half-cycle average: $(1/\pi)\int_0^\pi V_m \sin\theta \, d\theta = 2V_m/\pi$.

Memory trick: The sine's four numbers: **0.707 · 0.637 · 1.11 · 1.414** (rms, average, form, crest).

Exam trap: "220 V" is RMS; insulation must withstand the **peak 311 V**. Full-wave rectified sine has the **same** rms and average as a sine.

5 Phasors & Complex Numbers

Phasor: A complex number that carries the **magnitude and phase** of a sinusoid of a known frequency: $v = V_m \sin(\omega t + \varphi) \rightarrow \mathbf{V = V\angle\varphi}$ (V usually rms). Think of a vector rotating anticlockwise at ω; its vertical projection traces the sine. It turns calculus (d/dt → **jω**) into algebra.



rotating phasor and the sine it projects

Rectangular ↔ polar:

$a + jb = r\angle\theta$, $r = \sqrt{(a^2 + b^2)}$, $\theta = \tan^{-1}(b/a)$

$a = r\cos\theta$, $b = r\sin\theta$

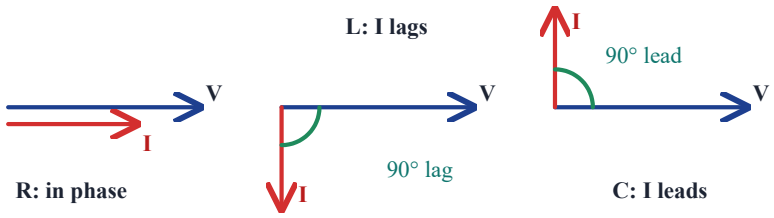
$\mathbf{j = \sqrt{-1} = 1\angle 90°}$, $j^2 = -1$, $1/j = -j$

Add/subtract in **rectangular**; multiply/divide in **polar** (multiply r, add angles)

Worked example 1 ✍️: $I = 100\angle 0° / (3 + j4) = 100\angle 0° / 5\angle 53.13° = \mathbf{20\angle -53.13°\ A} = 12 - j16$ A

Worked example 2 ✍️: $10\angle 0° + 10\angle 90° = 10 + j10 = \mathbf{14.14\angle 45°}$, not 20

Element	Time domain	Phasor law	Impedance Z	Phase of I w.r.t. V
R	v = Ri	V = RI	R	in phase (0°)
L	v = L di/dt	V = jωL I	$\mathbf{jX_L = j\omega L}$	I **lags** by 90°
C	i = C dv/dt	V = I/(jωC)	$\mathbf{-jX_C = 1/(j\omega C)}$	I **leads** by 90°



phasor diagrams with V as reference (anticlockwise = leading)

Memory trick: **ELI the ICE man**: in L, E (voltage) comes before I; in C, I comes before E. Or **CIVIL**: in C, I before V; V before I in L.

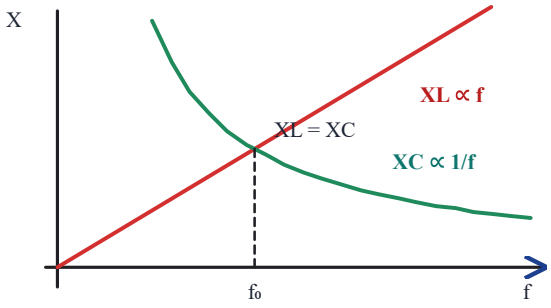
Exam trap: Phasors only combine sinusoids of the **same frequency**. Adding peak values of out-of-phase waves (10 + 10 = 20) is wrong: add them as complex numbers.

6 Impedance, Admittance & AC Networks

Impedance: $\mathbf{Z = V/I = R + jX = |Z|\angle\theta}$ (Ω). R = resistance, X = reactance. $|Z| = \sqrt{(R^2 + X^2)}$, $\theta = \tan^{-1}(X/R)$. X > 0 inductive, X < 0 capacitive.

Admittance: $\mathbf{Y = 1/Z = G + jB}$, unit **siemens (S)**. G = conductance, B = susceptance. $G = R/(R^2 + X^2)$, $B = -X/(R^2 + X^2)$.

	Inductor	Capacitor
Reactance	$\mathbf{X_L = 2\pi fL}$	$\mathbf{X_C = 1/(2\pi fC)}$
In Z	$+jX_L$	$-jX_C$
Susceptance	$B_L = -1/(\omega L)$	$B_C = +\omega C$
f → 0 (DC)	$X_L = 0$: short	$X_C \to \infty$: open
f → ∞	open	short
f doubled	X_L doubles	X_C halves



reactance against frequency

Network rules (same as DC, with Z): Series: $\mathbf{Z = Z_1 + Z_2 + \ldots}$ · Parallel: $\mathbf{Y = Y_1 + Y_2 + \ldots}$, two branches $Z = Z_1Z_2/(Z_1 + Z_2)$

Voltage divider: $\mathbf{V_1 = V \cdot Z_1/(Z_1 + Z_2)}$ · Current divider: $\mathbf{I_1 = I \cdot Z_2/(Z_1 + Z_2)} = I \cdot Y_1/(Y_1 + Y_2)$

Worked example 1: reactance ✍️: 0.1 H at 50 Hz: $X_L = 2\pi \times 50 \times 0.1 = \mathbf{31.4\ \Omega}$. 100 μF at 50 Hz: $X_C = 1/(2\pi \times 50 \times 10^{-4}) = \mathbf{31.8\ \Omega}$.

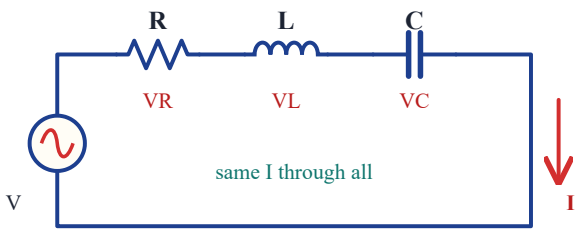
Worked example 2: parallel ✍️: R = 10 Ω ∥ X_C = 10 Ω on $100\angle 0°$ V. $Y = 0.1 + j0.1 = \mathbf{0.1414\angle 45°\ S}$; $Z = 1/Y = 7.07\angle -45° = \mathbf{5 - j5\ \Omega}$; $I = VY = \mathbf{14.14\angle 45°\ A}$ (leading).

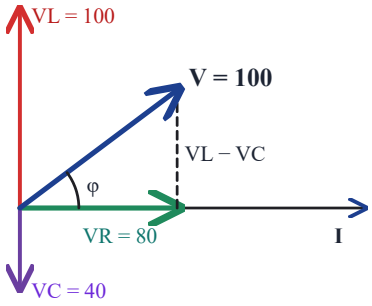
Worked example 3: divider ✍️: $100\angle 0°$ V across R = 6 Ω in series with $jX_L = j8$ Ω. $Z = 10\angle 53.13°$. $V_L = 100 \times j8/(6 + j8) = \mathbf{80\angle 36.87°\ V}$, $V_R = \mathbf{60\angle -53.13°\ V}$. Check: 60 + 80 = 140 ≠ 100, but $\sqrt{(60^2 + 80^2)} = \mathbf{100}$ ✓.

Exam trap: $G = 1/R$ **only** if the branch has no reactance; for R + jX, $G = R/(R^2 + X^2)$. Voltage magnitudes in AC **do not add** arithmetically.

Lesson 2 link: AC mesh and nodal analysis, Thevenin, Norton and superposition work exactly as in Lesson 2 (Nodal, Mesh & Circuit Theorems), using complex Z. Balanced Δ → Y: $\mathbf{Z_Y = Z_\Delta/3}$. AC maximum power transfer: $\mathbf{Z_L = Z_{th}^*}$ (complex conjugate), $P_{max} = |V_{th}|^2/(4R_{th})$.

7 Series RLC Circuit





Formulas: $Z = R + j(X_L - X_C)$ · $\mathbf{|Z| = \sqrt{(R^2 + (X_L - X_C)^2)}}$ · $\varphi = \tan^{-1}((X_L - X_C)/R)$ · $V = \sqrt{(V_R^2 + (V_L - V_C)^2)}$ · $pf = \cos\varphi = R/|Z|$

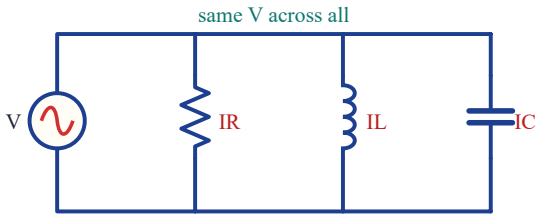
$X_L > X_C$: **inductive**, I lags · $X_L < X_C$: **capacitive**, I leads · $X_L = X_C$: **resonance**, Z = R

Worked example ✍️: R = 8 Ω, X_L = 10 Ω, X_C = 4 Ω on 100 V rms.

1. X = 10 − 4 = 6 Ω (inductive); Z = $\sqrt{(8^2 + 6^2)}$ = **10 Ω**, $\varphi = \tan^{-1}(6/8) = 36.87°$
2. I = 100/10 = **10 A**, lagging; V_R = 80 V, V_L = 100 V, V_C = 40 V; check $\sqrt{(80^2 + 60^2)} = 100$ ✓
3. pf = 8/10 = **0.8 lagging**; $P = I^2R$ = **800 W**, $Q = I^2X$ = 600 VAR, S = VI = 1000 VA

Exam trap: In series RLC the voltage across L or C can **equal or exceed the supply** (here V_L = 100 V = V). Only the phasor sum must equal V.

8 Parallel RLC Circuit



$Y = 1/R + j(\omega C - 1/\omega L)$ · $I = V \cdot Y$

$\mathbf{I = \sqrt{(I_R^2 + (I_C - I_L)^2)}}$

$I_L > I_C$ ($X_L < X_C$): inductive, lagging

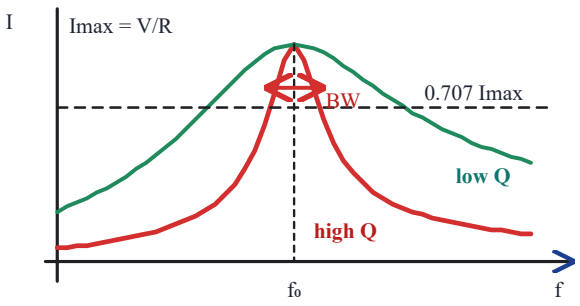
$I_C > I_L$: capacitive, leading

Worked example ✍️: R = 20 Ω, X_L = 10 Ω, X_C = 20 Ω in parallel on 100 V. I_R = 5 A, I_L = 10 A (−j), I_C = 5 A (+j). I = 5 − j10 + j5 = 5 − j5 = **7.07∠−45° A**; Z = 100/7.07 = **14.14 Ω**; pf = 0.707 lagging.

Memory trick: **Series: the bigger X wins. Parallel: the smaller X wins** (it draws the bigger current). Same $X_L < X_C$ → capacitive in series, inductive in parallel.

9 Resonance, Q-factor & Bandwidth

Resonance: The frequency at which $\mathbf{X_L = X_C}$, so the circuit is purely resistive (pf = 1, V and I in phase): $\boldsymbol{\omega_0 = 1/\sqrt{(LC)}}$, $\boldsymbol{f_0 = 1/(2\pi\sqrt{(LC)})}$. f_0 does not depend on R.



series resonance curve (half-power line dashed)

Quality factor (series): $\mathbf{Q = \omega_0 L/R = 1/(\omega_0 CR) = (1/R)\sqrt{(L/C)}}$

Bandwidth: $\mathbf{BW = f_0/Q} = R/(2\pi L)$ Hz

Half-power (−3 dB) points f_1, f_2: $I = I_{max}/\sqrt{2}$, $P = P_{max}/2$

$f_0 = \sqrt{(f_1 f_2)}$; for high Q, $f_{1,2} \approx f_0 \mp BW/2$

High Q → sharp peak, **better selectivity**

At resonance	Series RLC	Parallel (tank)
Impedance	**minimum**, Z = R	**maximum** (ideal: R; coil r: L/Cr)
Line current	**maximum**, V/R	**minimum**
Magnification	voltage: $V_L = V_C = QV$	current: $I_L = I_C = QI$
Q	$\omega_0 L/R$	$R/(\omega_0 L) = \omega_0 CR$
Name	**acceptor** circuit	**rejector** circuit
Below / above f_0	capacitive / inductive	inductive / capacitive

Practical tank (coil r + L, with C across): $f_0 = (1/2\pi)\sqrt{(1/(LC) - r^2/L^2)}$, slightly below $1/(2\pi\sqrt{LC})$; dynamic impedance $Z_D = L/(Cr)$.

Worked example ✍️: Series RLC: L = 10 mH, C = 1 µF, R = 10 Ω, V = 10 V.

1. $\omega_0 = 1/\sqrt{(10^{-2} \times 10^{-6})} = \mathbf{10^4}$ **rad/s**; $f_0 = 10^4/2\pi =$ **1.59 kHz** (not 10 kHz)
2. $X_L = X_C = \omega_0 L = 100\ \Omega$; $I_0 = V/R =$ **1 A**
3. $Q = 100/10 =$ **10**; $BW = 1591/10 =$ **159 Hz**; $V_L = V_C = Q \cdot V =$ **100 V** on a 10 V supply

Memory trick: **Series = Small Z, Sucks current in** (acceptor). **Parallel = Peak Z, Pushes current away** (rejector).

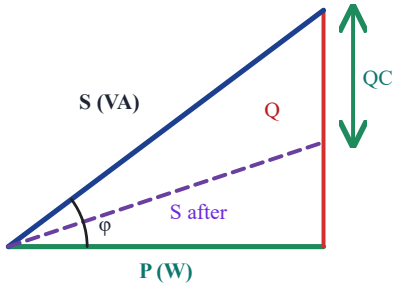
Exam trap: $f_0 \propto 1/\sqrt{(LC)}$: C × 4 → f_0 **halves**. Adding R changes Q and BW, never f_0.

IT link: Radio, Wi-Fi and mobile receivers tune by resonance; band-pass filters; quartz crystals (a very high-Q resonator) set CPU and RTC clocks; RFID/NFC tags resonate at 13.56 MHz.

10 AC Power & Power Factor

With $v = V_m \sin \omega t$ and $i = I_m \sin(\omega t - \varphi)$: $p = vi =$ **$VI \cos \varphi - VI \cos(2\omega t - \varphi)$** (rms V, I). The second term averages to zero (it pulses at **twice** the supply frequency), so the average power is $VI \cos \varphi$.

Power	Formula	Unit
Real / active P	**$VI \cos \varphi = I^2R$**	W
Reactive Q	$VI \sin \varphi = I^2X$	**VAR**
Apparent S	$VI = I^2\lvert Z\rvert$	**VA**
Complex S	$V \cdot I^* = P + jQ$	VA
Triangle	**$S^2 = P^2 + Q^2$**	



power triangle; a capacitor cuts Q

Power factor: **$pf = \cos \varphi = P/S = R/\lvert Z\rvert$**, between 0 and 1. **Lagging**: inductive load (motors, fans, transformers, ballasts), I lags V. **Leading**: capacitive load. Pure R: pf = 1, Q = 0. Pure L or C: pf = 0, P = 0. Inductive Q is positive (absorbed), capacitive Q negative.

Why low pf is bad: For the same P, $I = P/(V \cos \varphi)$ is larger → more I^2R line loss, thicker cables, bigger transformers, worse voltage drop. Utilities penalise industrial consumers with low pf.

Memory trick: **P does the work, Q sloshes back and forth, S is what the wires carry.** Units: **W**atts work, **VAR** varies, **VA** is the total.

pf correction: Connect a capacitor in parallel: **$Q_C = P(\tan \varphi_1 - \tan \varphi_2)$**, $C = Q_C/(\omega V^2)$. P is unchanged; S and line current fall.

Worked example ✍️: 10 kW load, pf 0.8 lagging, 220 V, 50 Hz; raise pf to 0.95.

1. $\tan \varphi_1 = 0.75$ ($Q_1 = 7.5$ kVAR, $S_1 = 12.5$ kVA); $\tan \varphi_2 = \tan(\cos^{-1} 0.95) = 0.329$
2. $Q_C = 10\,000 \times (0.75 - 0.329) =$ **4.21 kVAR**; $C = 4213/(314.16 \times 220^2) =$ **277 µF**
3. Line current $10\,000/(220 \times 0.8) = 56.8$ A → $10\,000/(220 \times 0.95) =$ **47.8 A** (−16%)

Exam trap: A wattmeter reads **P**, not VI. Generators, transformers and UPSs are rated in **VA** because their heating depends on current, whatever the pf.

IT link: A 1000 VA UPS at pf 0.6 supplies only **600 W**; server PSUs use active PFC (pf > 0.9). 1 unit = **1 kWh** = 3.6 MJ.

Night-Before-the-Exam Sheet

Formula sheet:

$i = C\,dv/dt$ · $W = \frac{1}{2}CV^2$

$v = L\,di/dt$ · $W = \frac{1}{2}LI^2$

C series: $1/C = \Sigma 1/C_k$ · parallel: ΣC_k

L series: ΣL_k · parallel: $1/L = \Sigma 1/L_k$

$\tau = RC$ · $\tau = L/R$ · $1\tau \rightarrow 63.2\%$

$x(t) = x(\infty) + [x(0) - x(\infty)]e^{-t/\tau}$

$V_{rms} = 0.707V_m$ · $V_{avg} = 0.637V_m$

FF = 1.11 · crest = 1.414 (sine)

$X_L = 2\pi fL$ · $X_C = 1/(2\pi fC)$

$Z = R + j(X_L - X_C)$ · $Y = G + jB$

$f_0 = 1/(2\pi\sqrt{LC})$ · $Q = \omega_0 L/R$

$BW = f_0/Q$ · $f_0 = \sqrt{(f_1 f_2)}$

$P = VI\cos\varphi$ · $Q = VI\sin\varphi$ · $S = VI$

$S^2 = P^2 + Q^2$ · $pf = P/S = R/|Z|$

$Q_C = P(\tan\varphi_1 - \tan\varphi_2)$

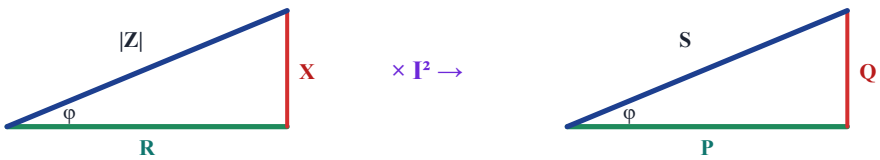
series $\alpha = R/2L$ · parallel $\alpha = 1/2RC$

Top 12 one-liners for MCQs

1. C blocks DC; L passes DC (short)
2. v_C and i_L cannot jump
3. BD mains: 220 V rms, 50 Hz, 311 V peak
4. Pure L: I lags 90° · pure C: I leads 90°
5. Unit of Y, G, B: siemens
6. Unit of Q: VAR · unit of S: VA
7. Series resonance: Z min, I max (acceptor)
8. Parallel resonance: Z max, I min (rejector)
9. At resonance pf = 1
10. Q ↑ → BW ↓ → selectivity ↑
11. Critical damping: $R = 2\sqrt{(L/C)}$ (series)
12. Low pf → more current, more loss

Top traps: $\tau = L/R$ not LR · series C carry equal charge (smallest C gets most V) · "220 V" is rms · AC voltages add as phasors · $G \neq 1/R$ in a series R-X branch · ω_0 in rad/s ≠ f_0 in Hz · series: bigger X wins, parallel: smaller X wins · f_0 does not depend on R · wattmeter reads P

All tricks together: L adds like R, C is R upside-down · 63-86-95-98-99, five taus and done · sine numbers 0.707, 0.637, 1.11, 1.414 · ELI the ICE man / CIVIL · Series Small Z (acceptor), Parallel Peak Z (rejector) · P works, Q sloshes, S is carried



impedance triangle × I^2 = power triangle (same angle φ)

✓ Practice MCQs (exam pattern, 10 marks)

1. Energy stored in a 10 μF capacitor charged to 100 V:

(a) 1 mJ (b) 0.05 J

(c) 0.5 J (d) 50 J

2. In DC steady state an ideal inductor acts as:

(a) a short circuit (b) an open circuit

(c) a resistor (d) a voltage source

3. Time constant of R = 1 MΩ, C = 2 μF:

(a) 0.5 s (b) 1 s

(c) 2 s (d) 20 s

4. Form factor of a sine wave:

(a) 1.414 (b) 1.0

(c) 1.57 (d) 1.11

5. In a purely capacitive circuit, the current:

(a) lags V by 90° (b) is in phase

(c) leads V by 90° (d) lags V by 180°

6. At series resonance the impedance is:

(a) minimum, = R (b) maximum

(c) zero (d) infinite

7. Power factor of a series circuit with R = 6 Ω, X = 8 Ω:

(a) 0.6 (b) 0.8

(c) 0.75 (d) 1

8. Unit of reactive power:

(a) W (b) VA

(c) VAR (d) J

9. In a series LC circuit C is made 4 times larger. f_0 becomes:

(a) $4f_0$ (b) $2f_0$

(c) $f_0/4$ (d) $f_0/2$

10. The unit of admittance is:

(a) ohm (b) siemens

(c) henry (d) farad

Answer key → 1-b · 2-a · 3-c · 4-d · 5-c · 6-a · 7-a · 8-c · 9-d · 10-b

Written question: v = 141.4 sin(1000t) V is applied to a series circuit of R = 30 Ω, L = 80 mH and C = 25 μF. Find X_L, X_C, Z, I, the power factor, P, Q, S, and the resonant frequency. Draw the phasor diagram. 10 marks

1. V = 141.4/√2 = **100 V** rms, ω = 1000 rad/s; $X_L = \omega L$ = **80 Ω**, $X_C = 1/(\omega C)$ = **40 Ω** 2
2. Z = 30 + j40 = **50∠53.13° Ω** (inductive); I = 100/50 = **2 A**, lagging 53.13° 2
3. pf = 30/50 = **0.6 lag**; P = I^2R = **120 W**, Q = I^2X = **160 VAR**, S = VI = **200 VA** ✓ 3
4. V_R = 60 V, V_L = 160 V, V_C = 80 V; phasor diagram as on p. 7 1
5. $\omega_0 = 1/\sqrt{(0.08 \times 25\ \mu)}$ = 707.1 rad/s → **f_0 = 112.5 Hz** < f = 159.2 Hz, so inductive ✓ 2

Before the exam ✓

- ✓ Can write the C/L duality table
- ✓ Can solve an RC/RL switching problem
- ✓ Know rms, average, FF, crest of 4 waves
- ✓ Can draw series RLC phasor diagram
- ✓ Can find f_0, Q, BW and size a pf capacitor

Also practise (written):

① Derive W = ½CV²

② Derive RC charging

③ Prove $V_{rms} = V_m/\sqrt{2}$

④ Series vs parallel resonance